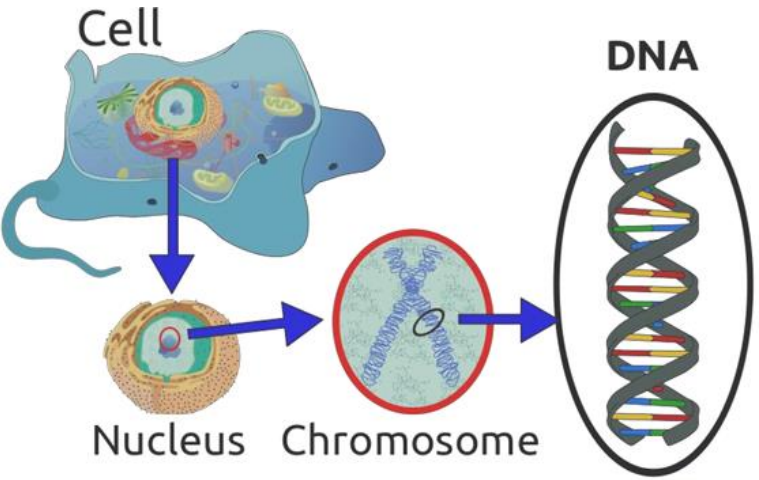
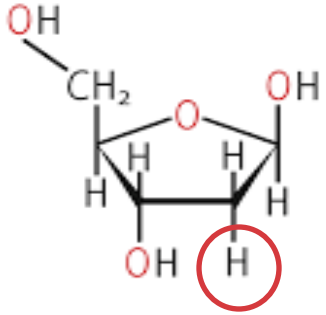
Nucleic Acids

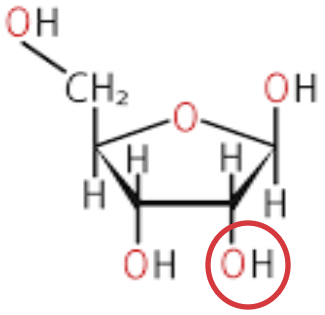


- Biologists suspected **transmission of genetic information** occurs in the **nucleus** in coiled structures called **chromosomes** (Boveri-Sutton 1902).
- Humans have 46 chromosomes (23 pairs) each made of genes. A **gene** is the **portion of DNA** responsible for the **synthesis of a specific protein**.
- **Nucleic acids** are polymers composed of repeating **nucleotide** monomers that **store and transfer genetic information**.

Deoxyribonucleic Acid (DNA):

- Monosaccharide D-**2**-deoxyribose
- Located in cell nucleus
- Stores /transfers genetic info
- Passed to new cells during cell division



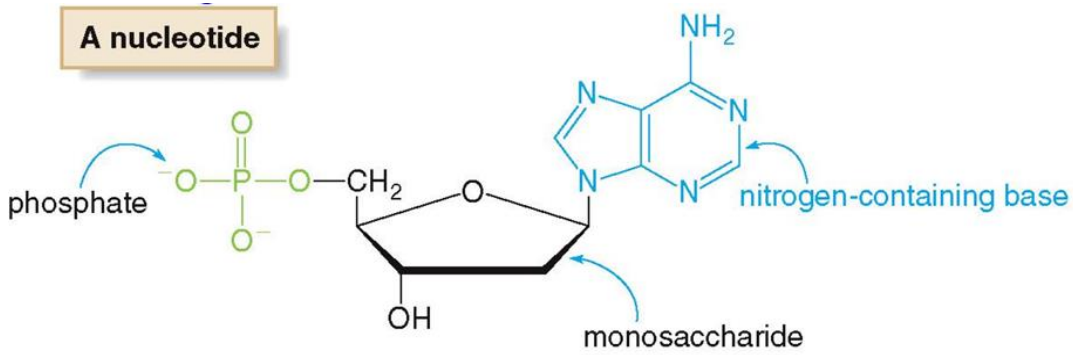


Ribonucleic Acid (RNA):

- Monosaccharide D-ribose
- Found in all parts of a cell
- Translates genetic information contained in DNA into proteins

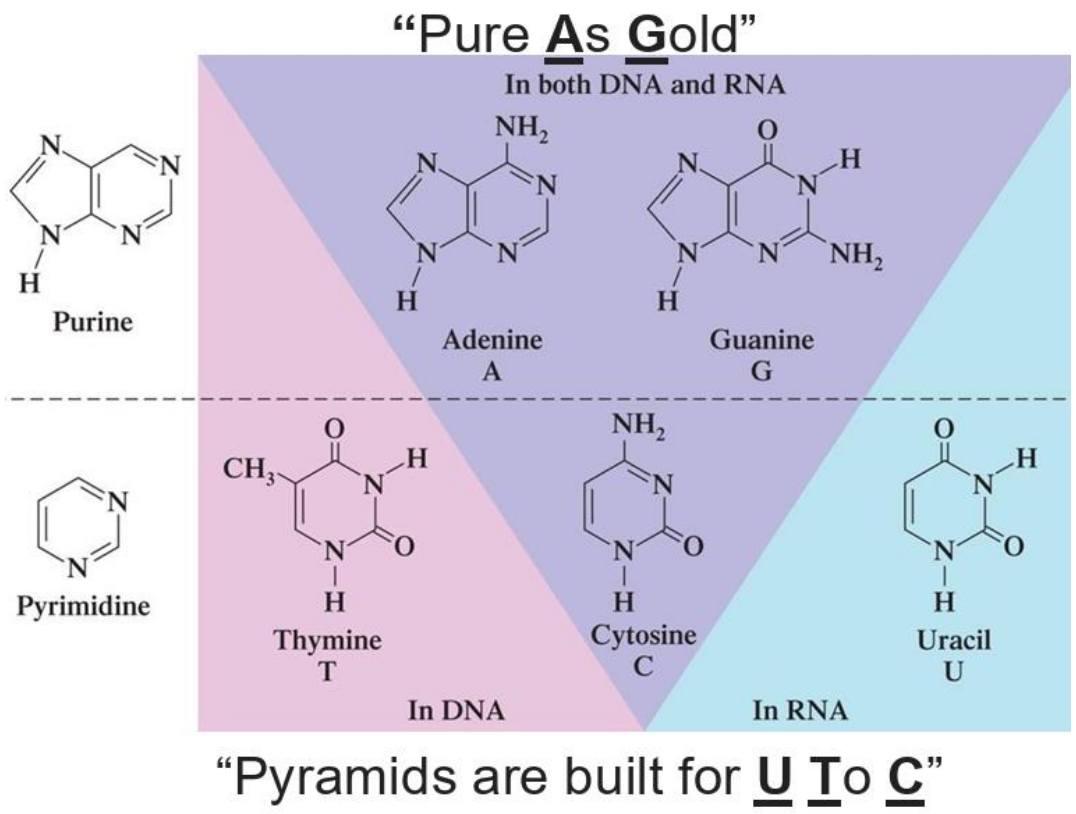
General Structure

- Nucleo**s**ides contain a base and a sugar
- Nucleo**t**ides contain a base, a sugar, and a phosphate (nucleoside + phosphate)
- Nucle**ic** Acids are polymers of nucleotides



Component	DNA	RNA
Bases	A, G, C, and T	A, G, C, and U
Sugar	Deoxyribose	Ribose
Nucleoside	Base + deoxyribose	Base + ribose
Nucleotide	Base + deoxyribose + phosphate	Base + ribose + phosphate
Nucleic Acid	Polymer of deoxyribose nucleotides	Polymer of ribose nucleotides

- N-containing base in each nucleotide is one of 5 types (based on purine or pyrimidine)
- Both DNA and RNA contain AGC;
  Only DNA contains T; Only RNA contains U

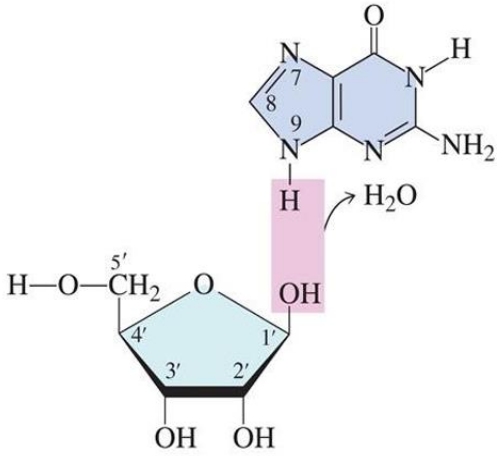


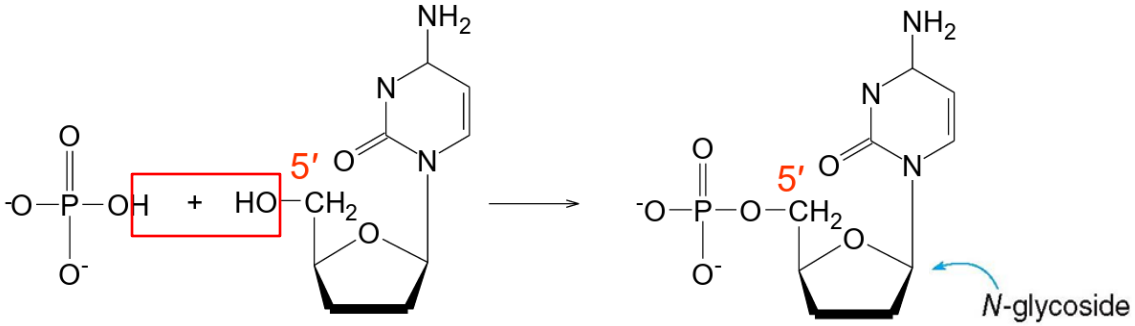
Formation & Naming

Formation:

Nucleo**s**ide: OH on anomeric C of sugar condenses with NH of base (β-N-glycosidic bond)

Nucleo**t**ide: OH on C**5′** of sugar in a nucleoside condenses with HPO_4^{2-}





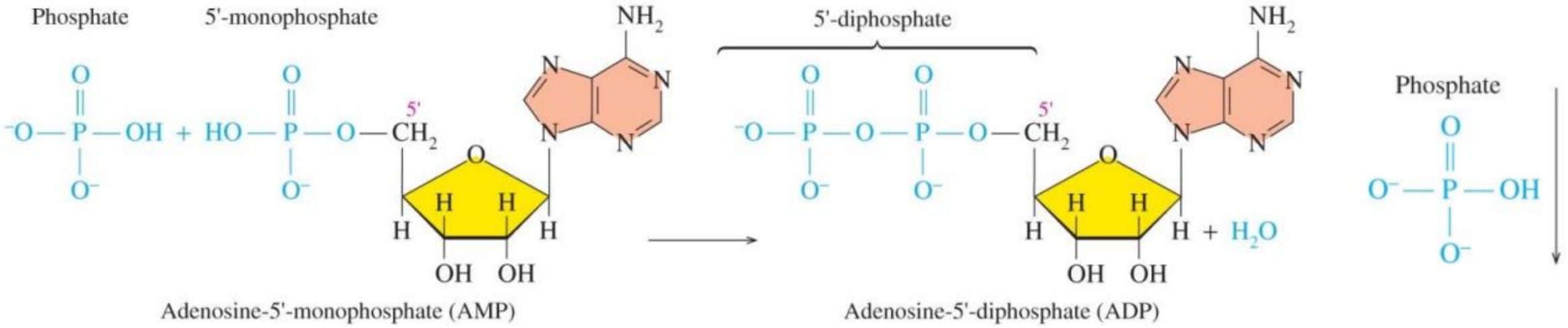
Base	Nucleosides	Nucleotides
RNA		
Adenine (A)	Adenosine (A)	Adenosine-5′-monophosphate (AMP)
Guanine (G)	Guanosine (G)	Guanosine-5′-monophosphate (GMP)
Cytosine (C)	Cytidine (C)	Cytidine-5′-monophosphate (CMP)
Uracil (U)	Uridine (U)	Uridine-5′-monophosphate (UMP)
DNA		
Adenine (A)	Deoxyadenosine (A)	Deoxyadenosine-5′-monophosphate (dAMP)
Guanine (G)	Deoxyguanosine (G)	Deoxyguanosine-5′-monophosphate (dGMP)
Cytosine (C)	Deoxycytidine (C)	Deoxycytidine-5′-monophosphate (dCMP)
Thymine (T)	Deoxythymidine (T)	Deoxythymidine-5′-monophosphate (dTMP)

Naming:

- Nucleoside from pyrimidine, use suffix "-idine"
- Nucleoside from purine, use suffix "-osine"
- Deoxyribonucleosides, add the prefix "deoxy-"
- Nucleotides: add "**5′**-monophosphate" to the end

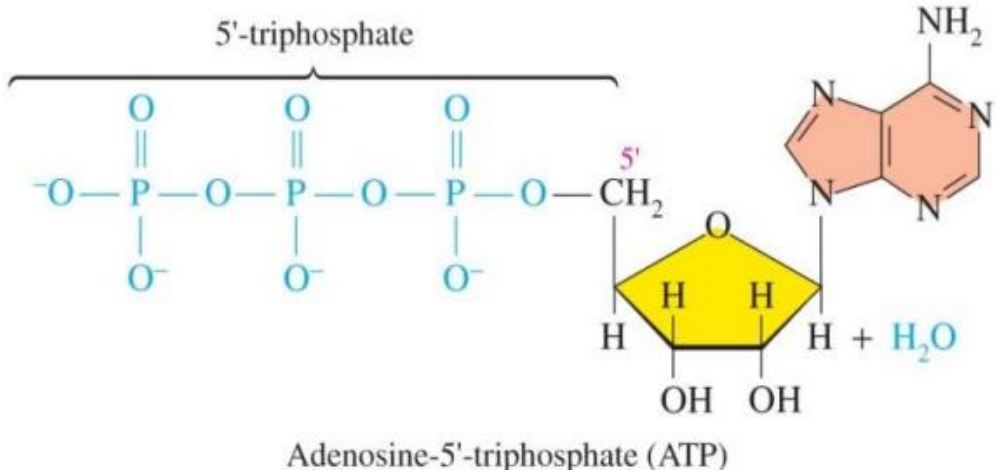
C atoms are usually numbered with primes (′) to distinguish them from the atoms in the bases.

AMP, ADP, ATP



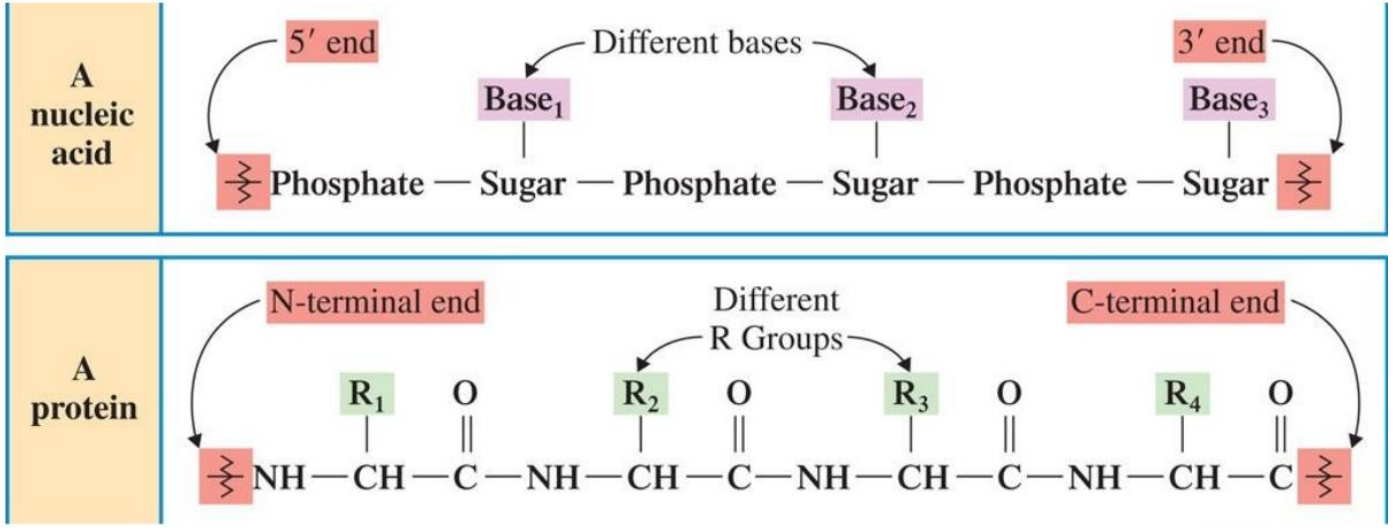
Adding phosphate groups to **AMP** forms the diphosphate **ADP** and the triphosphate **ATP**.

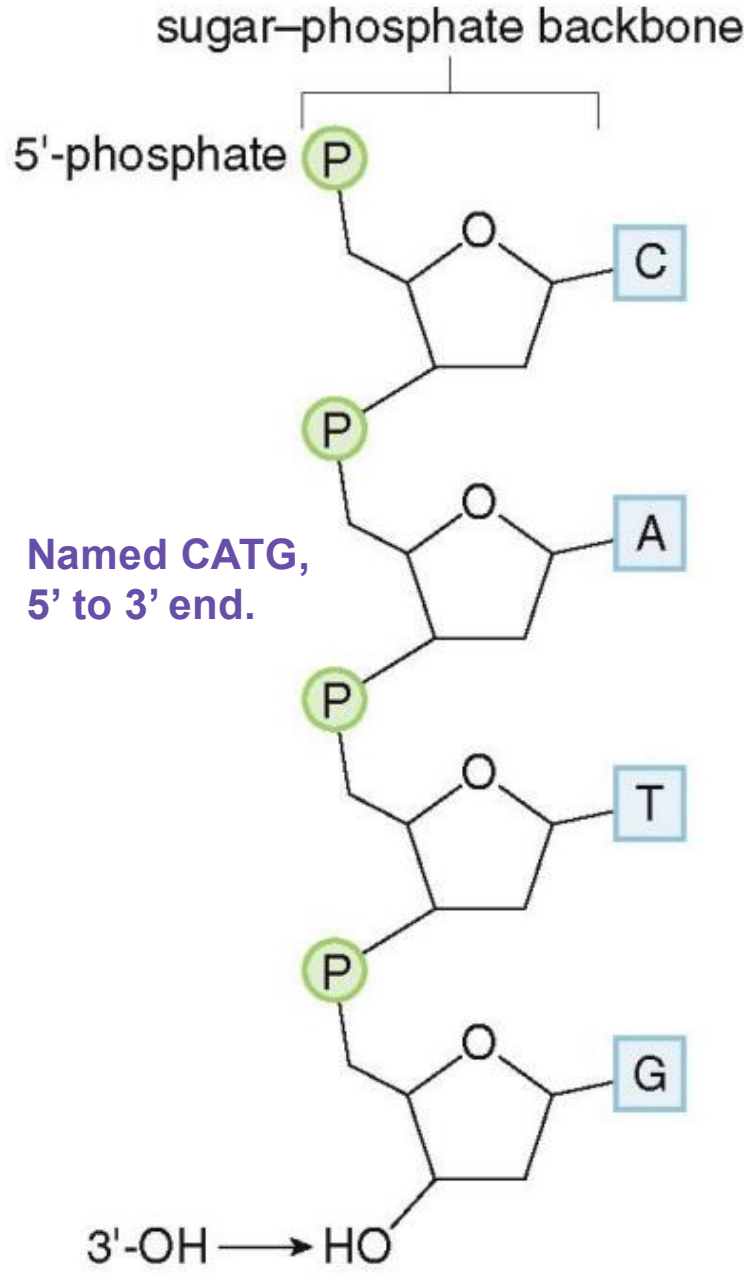
The cycle of **phosphorylation** and **hydrolysis** between these structures is essential to cell **metabolism, energy production, and storage**.



Nucleic Acid Primary Structure

- Polymers of nucleotides joined by phosphodiester linkages.
- Backbone consisting of **alternating sugar and phosphate** groups. **Phosphates link 5′ to 3′ carbon.**
- The **identity and order of the bases** distinguish one **polynucleotide** from another – **primary structure**.
- **Free phosphate** group at **5′ end** and **free OH** group at **3′ end**
- In DNA, the sequence of the bases carries genetic information.

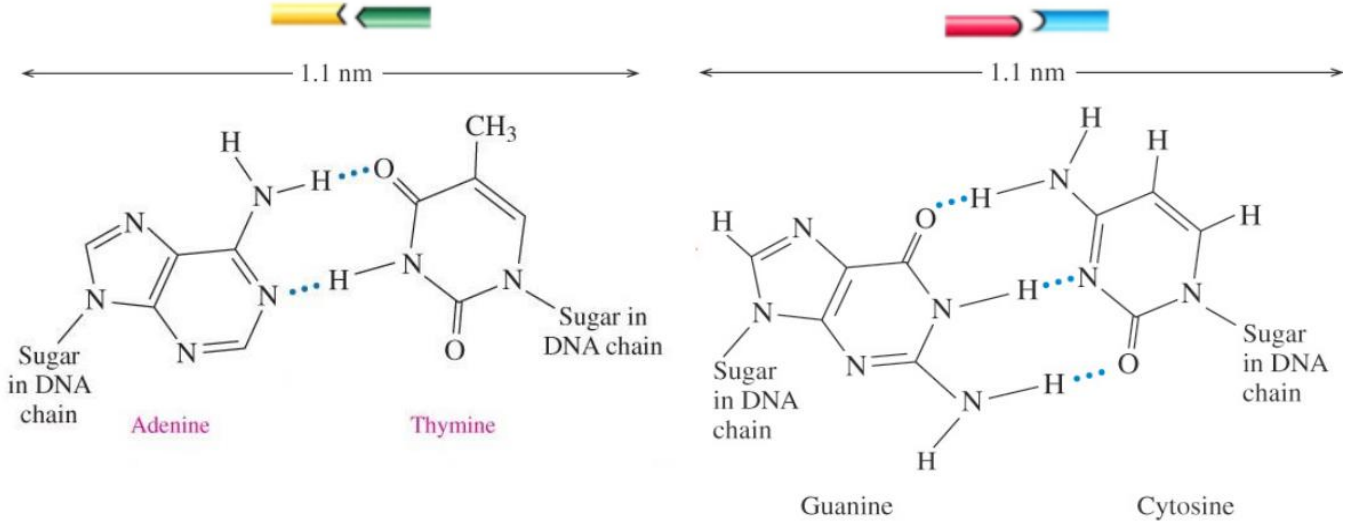




Named CATG, 5’ to 3’ end.

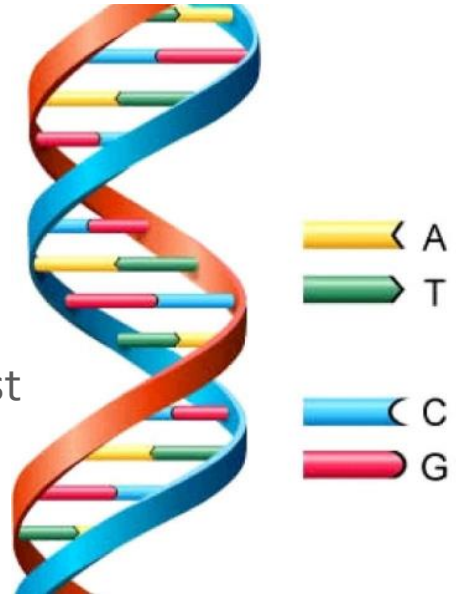
Nucleic Acid Secondary Structure

- Amounts of A,T,G, and C present in DNA molecules helped determine three-dimensional structure. **Chargaff's Rules:**
  - amounts of A and T are always equal
  - amounts of C and G are always equal
- **DNA** consists of **two polynucleotide strands** that wind into a **righthanded, anti-parallel double helix shape**.
  - Sugar-phosphate groups are outside, bases are inside
  - Held together by hydrogen bonds



Out of 6 combos, 2 are favorable

Size of interior limits pairs
- 2 purines = too big
- 2 pyrimidines = too small
- Purine and pyrimidine = best

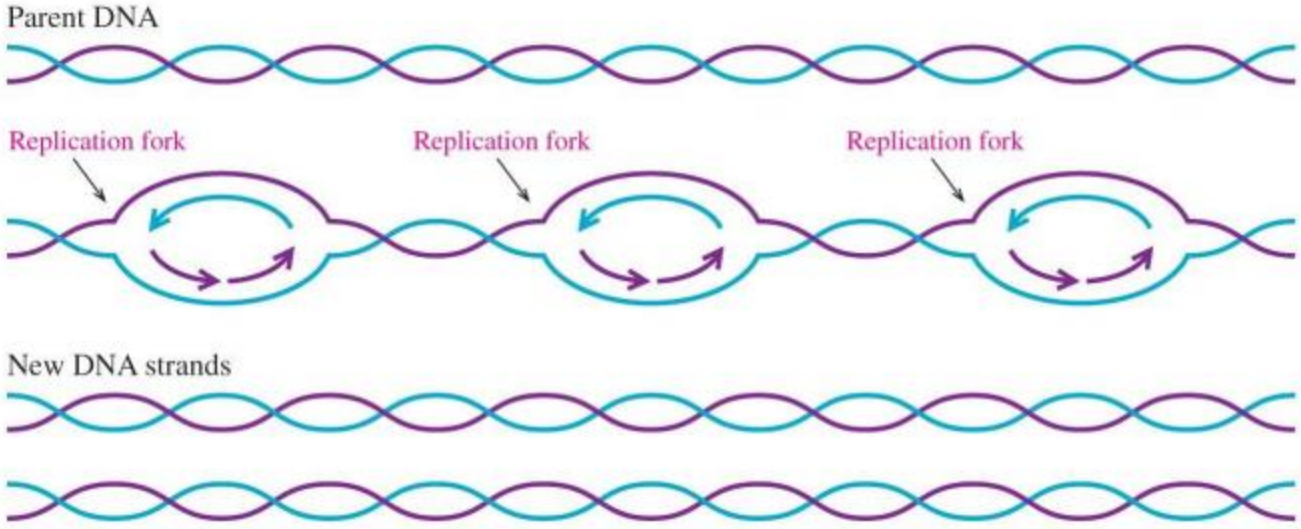


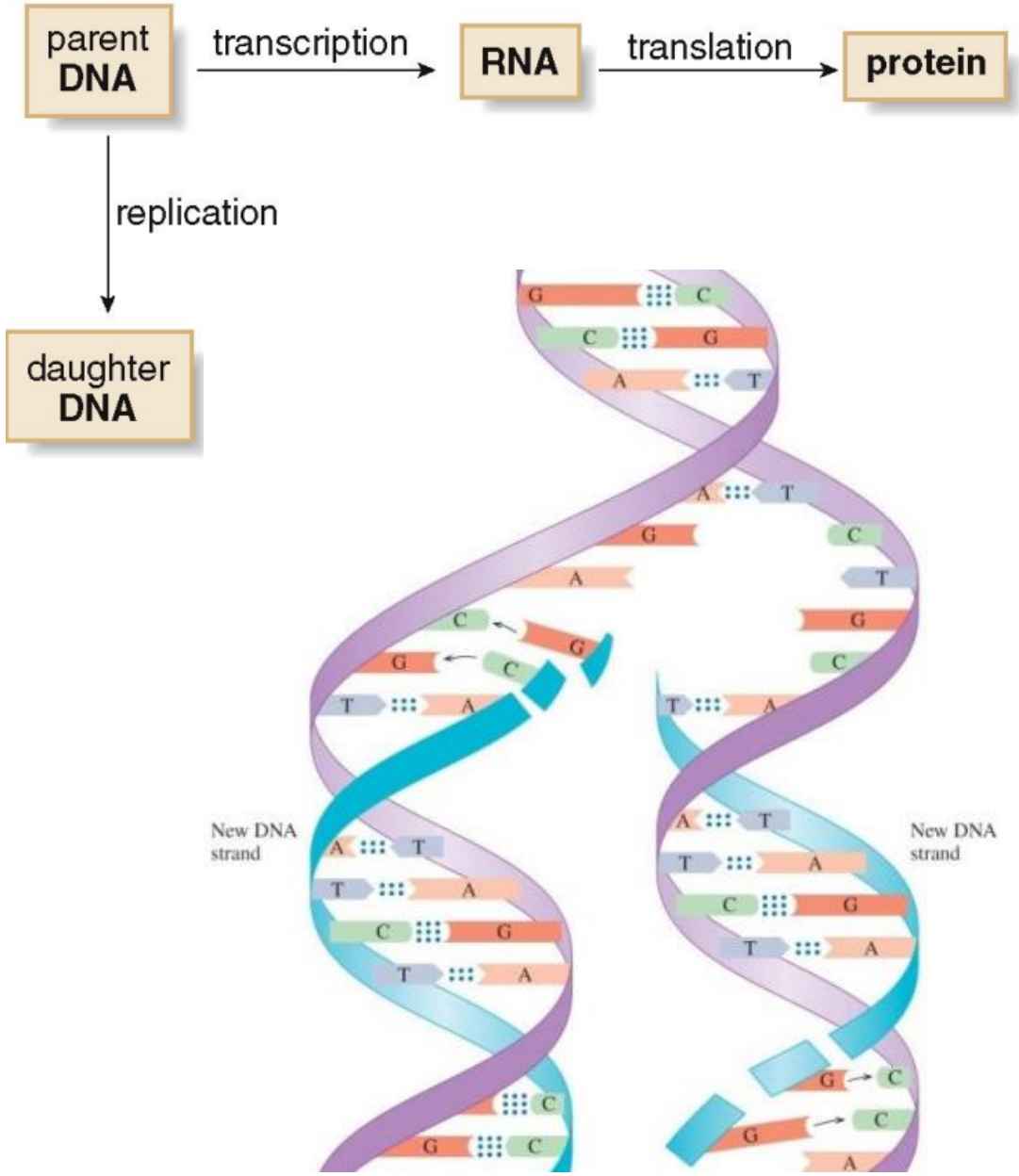
Write the complementary base sequence for the strand of DNA section shown :

5′—G—A—T—T—A—C—A—T—C—3′

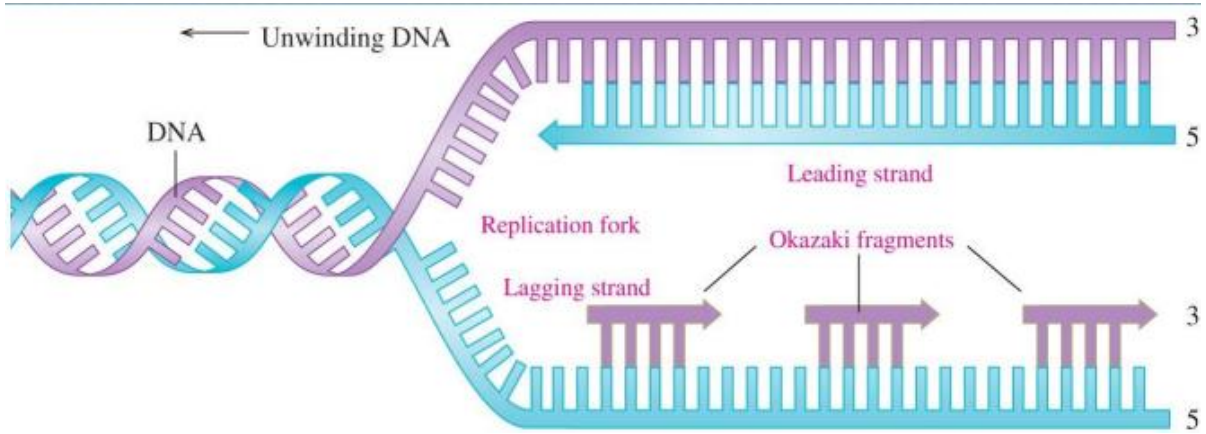
DNA Replication

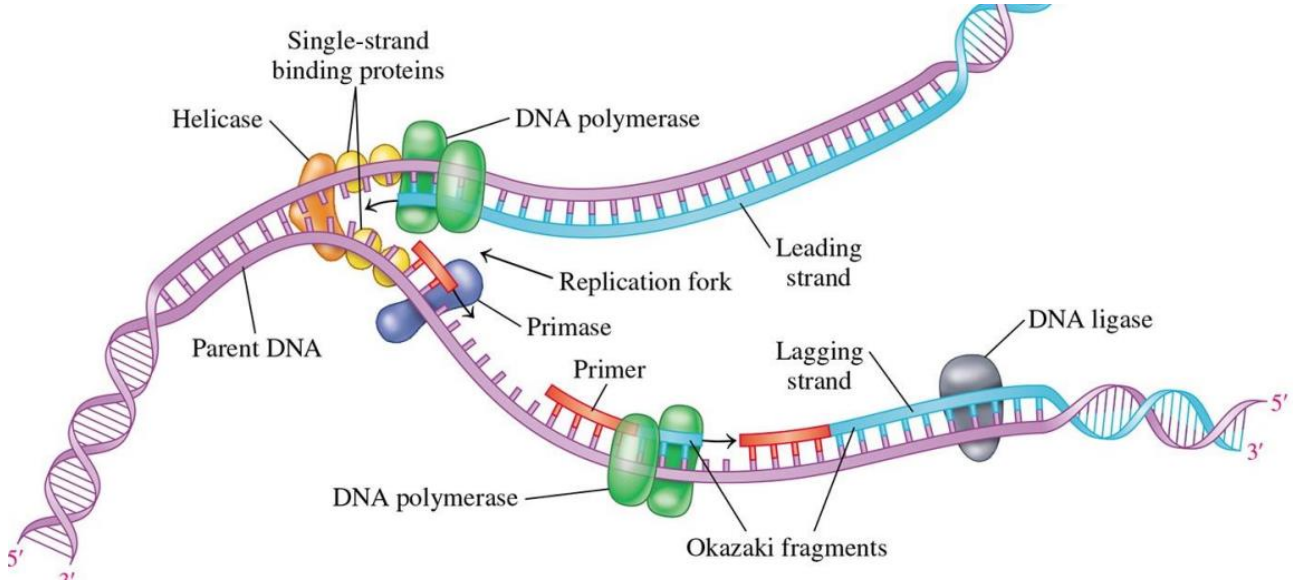
- Each time a **cell divides**, an **exact copy** of the **DNA** of the **parent** cell is needed for the new **daughter** cell.
- DNA **replication** involves
  1. **unwinding** the parent DNA;
  2. using each half as a **template**;
  3. **pairing** the bases in each strand with new bases to form new complementary daughter strands;
  4. daughter strands winding together with parent strands to form two **identical copies** of DNA.

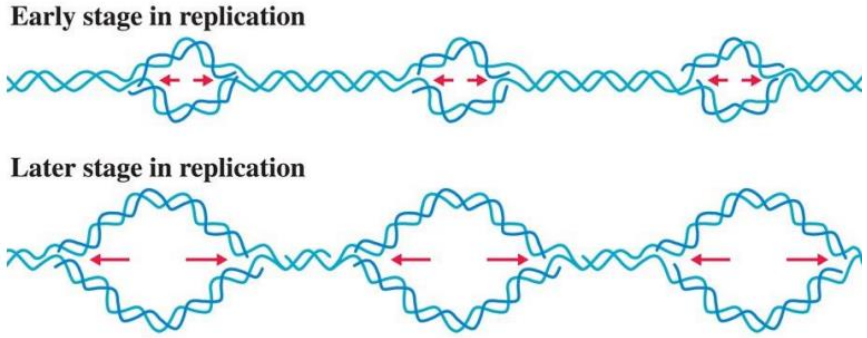




DNA Replication







- DNA replication usually occurs **bi-directionally** at **multiple sites**
- An enzyme, ***helicase***, unwinds the parent DNA at several sections
- At each open DNA section (**replication fork**), DNA ***polymerase*** catalyzes **5′–3′** ester bond formation in the **leading strand**
- **Lagging strand** (**3′–5′** direction) made in short sections called **Okazaki fragments**
- DNA ***ligase*** joins the Okazaki fragments to give a **single** 3′–5′ DNA strand
- **Semi-conservative process**: each new DNA double helix contains one strand of parent